SYNTHETIC AUTOSAR HIGH-LEVEL DESIGN: POWERTRAIN BATTERY MANAGEMENT SYSTEM (BMS)

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1. Document Metadata & System Context

Document ID: HLD-SYN-BMS-001 | Revision: 2.1 | Classification: Synthetic Open Engineering Knowledge
The High-Voltage Battery Management System (BMS) supervisory software manages 800V lithium-ion pack cells, high-voltage contactors, and thermal conditioning pumps. It executes within an AUTOSAR Classic 4.4.0 Runtime Environment (RTE).

2. High-Level Architectural Objectives

The BMS architecture isolates hardware-specific drivers through the MCAL layer, executes State of Charge (SOC) estimation, monitors per-cell voltage via isolated SPI buses, and controls contactor isolation relays under hard real-time constraints.

Subsystem	AUTOSAR Layer	Execution Periodicity	Safety ASIL Level
BMS Cell Supervisor	Application SWC	10 ms cyclic	ASIL C
BMS Thermal Control	Application SWC	100 ms cyclic	ASIL B
BMS Contactor Manager	Complex Driver (CDD)	5 ms cyclic	ASIL D
BMS State Estimator	Application SWC	20 ms cyclic	ASIL C

3. Software Components (SWC) and Interface Catalog

Component Name: BmsCellSupervisor_SWC
Type: Application Software Component
Description: Ingests digitized cell voltages and thermistor readings from the Cell Monitoring Circuit (CMC). Evaluates over-voltage (OV threshold: 4.25V) and under-voltage (UV threshold: 2.70V).
Ports:
- **RPort_CellVoltage** (ReceiverPort): Interface *If_CellVoltageArray* (Sender-Receiver), receives 192 cell voltage channels.
- **PPort_CellAlert** (ProvidePort): Interface *If_CellFaultNotification* (Client-Server), notifies Contactor Manager upon thermal runaway or cell voltage threshold breach.

Component Name: BmsThermalControl_SWC
Type: Application Software Component
Description: Controls coolant pump duty cycle and PTC cabin-chiller valves based on pack peak temperature readings.
Ports:
- **RPort_PackTemp**: Interface *If_TemperatureDistribution*.
- **PPort_PumpCmd**: Interface *If_CoolantActuator* (Sender-Receiver, 0-100% PWM).

Component Name: BmsSocEstimator_SWC
Description: Implements Extended Kalman Filter (EKF) combining coulomb counting and open-circuit voltage curves.
Ports:
- **RPort_CurrentSensor**: Interface *If_PackCurrent* (sampled at 1 kHz).
- **PPort_EstimatedSoc**: Interface *If_StateOfCharge* (0.0% to 100.0% float32).

4. Functional Flows and Sequence Traceability

Sequence FF-BMS-01: Critical Over-Voltage Protection Flow
Step 1: Cell Monitoring ASICs sample pack module voltages every 10ms.
Step 2: *BmsCellSupervisor_SWC* detects any individual cell exceeding 4.28V for >3 consecutive cycles.
Step 3: *BmsCellSupervisor_SWC* asserts synchronous fault flag via *PPort_CellAlert*.
Step 4: *BmsContactorManager_CDD* issues emergency opening request to pyrofuse and main positive contactor within 20 milliseconds.
Step 5: Diagnostic Trouble Code (DTC) 0x1A094F is logged in Non-Volatile Memory (NVM).

5. Safety and Real-Time Timing Constraints

- **Safety Goal SG-01:** Prevent thermal propagation resulting from high-voltage overcharge (ASIL C). Fault Handling Time Interval (FHTI): 50 milliseconds.
- **Bus Latency Constraint:** CAN-FD bus payload transmission latency for pack voltage broadcast must not exceed 8 milliseconds.
- **Memory Constraints:** BmsCellSupervisor RAM footprint must stay under 32 KB; ROM text section allocation maximum 128 KB.